INCH-POUND
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DETAIL SPECIFICATION

REFERENCE CELL, SILVER-SILVER CHLORIDE – IMPRESSED CURRENT CATHODIC PROTECTION (ICCP)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, fabrication, and testing of silver-silver chloride (Ag/AgCl) reference cells. The reference cell, also known as a reference electrode, is a half cell with very consistent reproducible potential (voltage) against which the potentials of other metals are measured. The potential measurement is provided as a voltage signal to an impressed current cathodic protection (ICCP) system controller and may be used to control and monitor the ICCP system operation.

1.2 Classification. Ag/AgCl reference cells are of the following types, grades, classes, and performance levels, as specified (see 6.2).

1.2.1 Type.

a. Type I – Standard cell

b. Type II – Diver replaceable cell

1.2.2 Grade.

a. Grade A – Mesh reference cell

b. Grade B – Plate reference cell

1.2.3 Class.

a. Class I – Surface ship

b. Class II – Submarine

1.2.4 Performance level.

a. Performance A – Standard performance

b. Performance B – High performance

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Part or identifying number (PIN). PINs to be used for Ag/AgCl reference cells acquired to this specification are created as follows:

M	23919	-	X	X	X	X
Prefix for Military Specification	Specification Number	-	Type (see code below)	Grade (see code below)	Class (see code below)	Performance (see code below)

Type code		Grade code		Class code		Performance code	
Type	**Code**	**Grade**	**Code**	**Class**	**Code**	**Performance**	**Code**
I	1	A	A	I	1	A	A
II	2	B	B	II	2	B	B
Example: M23919-1A1A – Standard cell, mesh reference cell, surface ship, standard performance							

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1141 - Standard Practice for the Preparation of Substitute Ocean Water

ASTM E11 - Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

(Copies of these documents are available online at www.astm.org.)

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SAE INTERNATIONAL

SAE-AMS7276	-	Rubber: Fluorocarbon (FKM) High-Temperature-Fluid Resistant Low Compression Set For Seals In Fuel Systems and Specific Engine Oil Systems
SAE-AMS-P-83461	-	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275 °F (135 °C)
SAE-AS8660	-	Silicone Compound NATO Code Number S-736

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. All reference cells furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction.

3.3.1 Ag/AgCl electrode element.

3.3.1.1 Chemical purity. The silver (Ag) used in fabricating the mesh/plate shall contain not less than 99.95 percent silver metal base purity. The silver chloride (AgCl) coating shall also have a metal basis purity of not less than 99.95 percent (see 4.4.1).

3.3.1.2 Cleaning. The silver surface shall be degreased and surface oxide fully reduced (cleaned) by means of a hydrogen flame or equivalent chemical method just prior to application of AgCl. The silver surface shall be bright and shiny once it has been cleaned (see 4.4.2). The silver shall be handled with stainless steel tongs to prevent contamination.

3.3.1.3 Application of AgCl coating. The AgCl coating shall be applied by either an electroplating or hot dipping process. Sintering is prohibited unless prior approval of the process has been granted by NAVSEA (see 6.2).

3.3.1.4 Surface area. The surface area of a grade A silver element shall have a mesh size of 40 or finer. For a grade B element, the final exposed surface area of the plate shall be not less than 0.491 square inches (3.18 square centimeters) or as approved by NAVSEA (see 4.4.3 and 6.2).

3.3.1.5 Service life. The service life of a reference cell shall be 12 years without degradation at input impedance of the reference cell monitoring circuit (see 4.4.4).

3.3.1.6 Workmanship. The reference cell element shall have a surface visually, without magnification, devoid of cracks or foreign matter (see 4.4.2).

3.3.2 Electrical connection. The pin shall be made out of a noble material. The pin shall be plated in 99.95 percent silver or gold metal base purity (see 4.4.1) and shall have a measured resistance of less than 0.1 ohm (see 4.4.5). The pin and connection shall be made out of the same materials.

3.3.3 Reference cell assembly. The assembly shall show no signs of damage. All parts shall be accounted for and in working order, ready to install (see 4.4.2).

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3.3.3.1 Material. The reference cell assembly shall be made of a non-conductive material to ensure that there is isolation (no shorting or grounding) between the reference cell and the hull; polyvinyl chloride is recommended.

3.3.3.2 Cable. The cable shall show no signs of damage or oxidation when examined as specified in 4.4.2.

3.3.3.3 O-rings. O-rings for the hub of the base assembly shall be circular rubber rings in accordance with SAE-AMS-P-83461 or SAE-AMS7276. Sealing grease for installation of the O-rings shall be in accordance with SAE-AS8660.

3.3.3.4 Fasteners. The fasteners used to attach the assembly to the structure to be protected shall be made from corrosion-resistant material.

3.4 Performance requirements.

3.4.1 Hydrostatic pressure. Class I and II reference cells and assemblies shall be tested to withstand specified design pressure based on the functional platform requirements identified in 6.2. In the absence of specific pressures identified in 6.2, the hydrostatic test methodology of 4.4.6 shall be utilized. Product form reference cell elements, consistent with their performance level, shall maintain the accuracy requirements provided in 3.4.4 when subjected to the hydrostatic test methodology of 4.4.6. Reference cell element assemblies shall show no evidence of mechanical damage, water leakage, or impaired electrical characteristics (see 4.4.2).

3.4.2 Insulation resistance. The reference cell assembly insulation resistance shall be 1.0 megohm or greater versus the element cable/pin (see 4.4.7).

3.4.3 Shock. The reference cell assembly shall meet the grade B shock requirements in accordance with MIL-DTL-901 (see 4.4.8) without leakage when subjected to the applicable hydrostatic pressure (see 3.4.1).

3.4.4 Accuracy and offset. Performance A reference cells shall have a direct current (DC) voltage (potential) measurement within ±10 millivolts (mV) DC (see 4.4.9) as compared to the NAVSEA-approved standard reference cell. Performance B reference cells shall have a DC voltage (potential) measurement within ±4 mV DC (see 4.4.9 and 4.4.10.2) as compared to the NAVSEA-approved standard reference cell (see 4.4.9.2). Additionally, the difference between the maximum and minimum potential values of all performance B reference cells being tested shall not exceed 2 mV DC (see 4.4.9 and 4.4.10.2) and 8 mV DC during hydrostatic testing (see 4.4.6).

NOTE: For the remainder of this specification, the NAVSEA-approved standard reference cell shall be referred to as the Standard.

3.4.5 Static drift (performance B reference cells only). The maximum change in static drift (see 6.4.4) between any two reference cell elements shall not exceed 0.6 mV DC under constant environmental conditions over 14 days (see 4.4.10.3).

3.4.6 Temperature linearity (performance B reference cells only). The maximum absolute difference in slope between any two reference cell elements shall not exceed 0.035 mV DC/°F when tested at temperatures from 40 to 90 °F (4.5 to 32.2 °C) over 336 hours (see 4.4.10.5). The R-squared value (with a minimum of 20 data points) for a linear fit shall meet or exceed a value of 0.9800.

3.4.7 Water resistivity linearity (performance B reference cells only). The maximum absolute difference in slope, when plotted on a semi-log graph, between any two reference cell elements shall not exceed 0.03 mV DC/log(ohm-centimeter) when tested at water resistivities between 18 ohm-centimeters and 288 ohm-centimeters (see 4.4.10.6). The R-squared value (with a minimum of 20 data points) for a linear fit shall meet or exceed a value of 0.9800.

3.4.8 Underwater replaceable reference cells. Type II reference cells shall be interchangeable underwater by divers. DC potential measurements for performance A replacement reference cells shall be ±10 mV DC from the original product cell; DC potential measurements for performance B replacement reference cells shall be ±2 mV DC from the original product cell. Type II reference cells shall be constructed (see 3.3.3) such that the connection to the pin remains free from oxidation after underwater replacement (see 4.4.11).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be performed on reference cells and shall be as specified in table I. The sample size shall consist of 12 reference cells randomly selected from three electroplating baths (four reference cells selected per bath). All unique configurations shall be sampled.

4.2.1 Periodic retention of qualification. Retention of qualification shall be performed at intervals of not more than 5 years after initial qualification and shall be as specified in table I.

4.2.2 Change approval. A change in material, production processes, or production equipment used in the manufacture of reference cells, which have been qualified, shall require written approval by NAVSEA. Incorporation of any changes, which have not been so approved, shall require requalification of the item in question.

4.3 Conformance inspection. Conformance inspection shall be performed on all reference cells and shall be as specified in table I.

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TABLE I. Summary of requirements and tests.

Requirement paragraph	Requirement	Qualification inspection	Retention of qualification	Conformance inspection	Test paragraph
3.3.1	Ag/AgCl electrode element	X	X	X	4.4.1 through 4.4.4
3.3.2	Electrical connection	X	X	X	4.4.1 and 4.4.5
3.3.3	Reference cell assembly	X	X	X	4.4.2, 4.4.6, and 4.4.7
3.4.1	Hydrostatic pressure - static	X	X	X	4.4.6.2.1 and 4.4.7
3.4.1	Hydrostatic pressure - cycling	X	X		4.4.6.2.2 and 4.4.7
3.4.2	Insulation resistance	X	X	X	4.4.7
3.4.3	Shock	X			4.4.8
3.4.4	Accuracy and offset	X	X	X	4.4.9 – Short-term [1/] 4.4.10.2 – Long-term [1/]
3.4.5	Static drift	X [2/]	X [2/]		4.4.10.3
3.4.6	Temperature linearity	X [2/]	X [2/]		4.4.10.5
3.4.7	Water resistivity linearity	X [2/]	X [2/]		4.4.10.6
3.4.8	Underwater replaceable reference cells	X			4.4.11

NOTES:

[1/] Long-term accuracy and offset test (see 4.4.10.2) is for qualification inspection and retention of qualification for performance B reference cells. Short-term accuracy and offset test (see 4.4.9) is for qualification inspection for performance A cells, conformance inspection for performance A and B cells, and retention of qualification for performance A reference cells.

[2/] Qualification test for performance B reference cells only.

4.4 Test procedures.

4.4.1 Chemical purity. When specified by the qualifying activity or otherwise (see 6.2), gold and silver material certification shall be provided.

4.4.2 Visual inspection. A non-destructive, visual inspection of the cable, assembly (if applicable), and reference element shall be performed. Upon completion of testing, if necessary, a destructive examination shall be performed to inspect the integrity of the reference cell (see 4.4.6.2.2).

4.4.3 Surface area. The surface area of a grade B reference cell shall be measured using laboratory-calibrated digital calipers or detail computer aided design software. The mesh size of a grade A reference cell shall be measured in accordance with ASTM E11.

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4.4.4 Service life. When specified by the qualifying activity or otherwise (see 6.2), certification of uniform thickness and longevity (life) calculations shall be provided to support service life requirements. The life shall be based on the weight of AgCl used in the reference cell. The minimum amount of AgCl per square centimeter on the reference cell surface shall be not less than the value given by:

$$Weight_{AgCl}(mg) = 3 \times LeakageCurrent\ (A) \times DesignLife\ (\text{sec}) \times 1.49(\frac{\text{mg}}{\text{A}-\text{sec}})$$

Where: the leakage current is given as the hull polarization value (e.g., 0.85 volt) divided by the input impedance of the potential measuring circuit (ohms). Note that the amount of AgCl on the reference cell face may typically be greater than this value to ensure stable readings. The calculated value is only a minimum.

4.4.5 Electrical resistance testing. The electrical resistance shall be measured using an ohmmeter with a resolution of 0.001 ohms and run at ambient temperature (see 3.3.2). The measurement shall be taken from the reference cell element connection to the end of the supplied cable length.

4.4.6 Hydrostatic pressure.

4.4.6.1 Class I reference cells. Each sample electrode shall be mounted in a pressure test fixture so that the face and mating surface of the reference cell assembly are covered. A pressure of 100±10 pounds per square inch (690±69 kilopascals [kPa]) shall be applied to the base, electrode, and pin assembly prior to assembly of the holder (see figure 1). The tube inserted over the electrode hub and O-rings shall be filled with a colored, translucent liquid "soapy water" containing a wetting agent (i.e., Windex® or equal). The pressure shall be maintained for at least 2 minutes. No bubbles shall appear. Additionally, no deformation shall be present on the reference electrode or assembly.

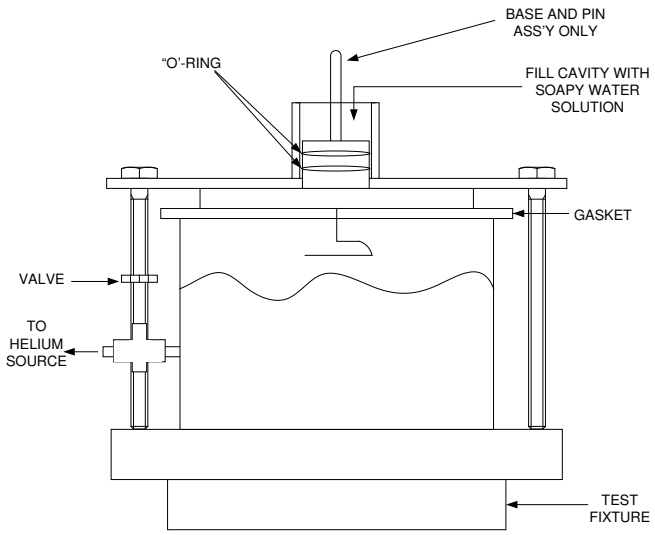


FIGURE 1. Sample pressure test setup for class I reference cells.

4.4.6.2 Class II reference cells. Reference cell pressures are governed by ship-specific criteria pertaining to the test depth of the vessel. In the absence of ship-specific criteria (see 6.2), the following criteria shall be used for hydrostatic and hydrostatic pressure cycling.

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4.4.6.2.1 Hydrostatic pressure – static. The reference cell element assembly, mating plug and cable segment attached, shall be subjected to the hydrostatic pressure test schedule of table II. After each step of table II, all pressure shall be removed before the next step is initiated. The test medium shall be seawater at a resistivity of 20 ±2 ohm-centimeters. The solution shall be allowed to equilibrate at ambient pressure and temperature for a minimum of 15 minutes prior to initiating testing. Insulation resistance shall be measured before and after the test. A minimum of three reference cell element assemblies shall be subjected to hydrostatic pressure testing within any given test. The reference cell element potentials shall be available for logging data throughout the test, and the differential potentials between all test cells shall be recorded at a minimum of once a minute with a minimum accuracy of 0.01 mV. The differential potentials measured shall meet the requirements of 3.4.4.

TABLE II. Hydrostatic pressure static test sequence.

Step	psig (kPa)	Hold time
1	20 (138)	5 Minutes
2	20 (138)	5 Minutes
3	20 (138)	5 Minutes
4	50 (435)	5 Minutes
5	130 (900)	5 Minutes
6	250 (1,725)	5 Minutes
7	500 (3,450)	5 Minutes
8	750 (5,175)	5 Minutes
9	1,000 (7,900)	24/2 Hours 1/

NOTE:

1/ Qualification testing (24 hours)/Conformance testing (2 hours)

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4.4.6.2.2 Hydrostatic pressure – cycling. Reference cell pressures are governed by ship-specific criteria pertaining to the test depth of the vessel. In the absence of specific test pressures, the reference cells, including a cell, mock steel baseplate, mating plug, and cable segment attached, shall be subjected to the hydrostatic pressure cycle testing schedule of table III. A temperature sensor shall be placed in the tank to monitor temperature changes throughout the test. The test medium shall be seawater at a resistivity of 20±2 ohm-centimeters. A fluorescent dye shall be added to the solution (recommend Kingscoat Chemicals Bright Dyes FWT Red 25 Liquid, part number 106023 or equivalent, which shall be used at a concentration of 1 pint of dye per 3,120 gallons of water or at the manufacturer's recommended concentration) for subsequent destructive testing to detect any defects in the equipment design. The solution shall be allowed to equilibrate at ambient pressure and temperature for a minimum of 15 minutes prior to initiating testing. Insulation resistance readings shall be taken before testing, after each seawater temperature change, and following the completion of the hydrostatic cycling test. A minimum of three reference cell assemblies shall be subjected to hydrostatic pressure cycle testing. The reference cell element potentials shall be available for logging data throughout the test, and the differential potentials between all test cells shall be recorded at a minimum of once a minute with a minimum accuracy of 0.01 mV. The differential potentials measured shall meet the requirements of 3.4.4. The test shall proceed as follows: 5 minutes at 0 pounds per square inch gauge (psig), 10 minutes to rise to test pressure (1,000 psig), 5 minutes hold at test pressure, and 10 minutes to drop to 0 pressure. The duration of each pressure cycle shall be 30±5 minutes. Seawater temperature ranges shall be 32 through 36 °F (0 through 2 °C) for the first 50 cycles, 40 through 70 °F (4 through 21 °C) for the next 100 cycles, and 75 through 90 °F (24 through 32 °C) for the final 50 cycles. When tested, the assembly shall withstand the fluorescent dye penetrant test as applicable without any clear leak path. Post testing, the test item shall be cleaned and disassembled. A minimum of two cuts shall be performed on the test item, one at the molded connector and another along the cable length. An ultraviolet light shall be used to visually inspect for fluorescent dye penetration into the molded cavities. Passing criteria for this test shall be no continuous path of fluorescent dye between the environment and the molded cavities.

TABLE III. Hydrostatic pressure cycle test sequence.

Step	Time	Beginning psig (kPa)	Pressure activity	End psig (kPa)	Temperature	Cycles
1	5 minutes	0 (0)	Hold	0 (0)	32 to 36 °F (0 to 2 °C)	50
2	10 minutes	0 (0)	Rise to	1,000 (7,900)		
3	5 minutes	1,000 (7,900)	Hold	1,000 (7,900)		
4	10 minutes	1,000 (7,900)	Fall to	0 (0)		
1	5 minutes	0 (0)	Hold	0 (0)	40 to 70 °F (4 to 21 °C)	100
2	10 minutes	0 (0)	Rise to	1,000 (7,900)		
3	5 minutes	1,000 (7,900)	Hold	1,000 (7,900)		
4	10 minutes	1,000 (7,900)	Fall to	0 (0)		
1	5 minutes	0 (0)	Hold	0 (0)	75 to 90 °F (24 to 32 °C)	50
2	10 minutes	0 (0)	Rise to	1,000 (7,900)		
3	5 minutes	1,000 (7,900)	Hold	1,000 (7,900)		
4	10 minutes	1,000 (7,900)	Fall to	0 (0)		

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4.4.7 Insulation resistance testing. Insulation resistance readings shall be taken before and after the hydrostatic test using a 50-volt megohmmeter. Insulation resistance shall be measured for three interfaces: cable jacket to the conductor, cable jacket to the shield layer, and conductor to the shield layer (see 3.4.2).

4.4.8 Shock testing. Grade B shock testing shall be accomplished in accordance with MIL-DTL-901. Subsequent to the shock test, the reference cell(s) shall pass a hydrostatic pressure test using the design pressure in accordance with 3.4.3 and the methodology of 4.4.6. When specified by the qualifying activity or otherwise (see 6.2), certification of test results shall be provided.

4.4.9 Short-term accuracy and offset testing. All product form performance A and B reference cell elements shall be tested for accuracy and offset compliance against the Standard (see 4.4.9.2). Each reference cell shall be tested in 0.6 normal (N) sodium chloride (NaCl) solution as prepared in 4.4.9.1 and at a constant temperature of 77±1.8 °F (25±1 °C). The measurements shall be made with a voltmeter with a resolution of 0.01 mV on the 100 mV scale and a minimum input resistance of 10 gigaohm.

4.4.9.1 Preparation of 0.6 N NaCl solution. Thirty-five grams (1.23 ounces [oz]) of NaCl crystals shall be measured (with a scale of minimum precision of ±0.1 milligrams) for each 1,000 milliliters (ml) (33.8 oz) of solution required. The NaCl shall be poured into a 1,000-ml (1-liter or 33.8 oz) volumetric flask, beaker, or any other container marked to contain 1,000 ml (33.8 oz). The container shall be filled to a volume of 1,000 ml (33.8 oz) with distilled water. The solution shall be stirred to dissolve all NaCl crystals.

4.4.9.2 Ag/AgCl standard cell. The sintered Ag/AgCl Standard shall have a minimum surface area of 0.0254 square inches (0.164 square centimeters). For a circular reference cell, this would be a 0.18-inch (4.5-millimeter) diameter. The Standards are EC-DC-AGE300LP, model number AGE300LP or AG-AGCL ELECTRODE DISC ELECTRODE12.5MM, model number 64-1316 (E2444) or equal as approved by NAVSEA.

4.4.9.3 Standard cell preparation and validation. A minimum of 10 Ag/AgCl standard reference cells (see 4.4.9.2) shall be procured for immersion with an epoxy-sealed electrical connection. All cells shall be placed in the 0.6 N NaCl solution as specified in 4.4.9.1, making sure to remove any air bubbles on the element surface. The standard reference cells shall be allowed to equilibrate for a minimum of 96 hours at room temperature. One of the cells shall be arbitrarily chosen as the base reference cell and the potential of all other nine cells versus this base cell shall be measured to an accuracy of ±0.01 mV DC. From the distribution obtained, three cells shall be chosen that are within ±0.5 mV from each other. These are termed the “standard electrodes” hereafter. One of the three standard electrodes shall be arbitrarily chosen as the “Working Standard Electrode”; this is the standard cell that all product form electrodes shall be measured against. The remaining two standard electrodes (termed “Control Standard Electrodes” hereafter) shall be utilized to assure conformance of the Working Standard Electrode. A voltmeter shall be used as specified in 4.4.9 to measure the differential potential between the Working Standard Electrode and each Control Standard Electrode. In addition, the differential potential between each Control Standard Electrode shall be measured. If the ±0.5 mV criterion is exceeded, each of the cell(s) causing the discrepancy shall be replaced with a new one of the same type.

4.4.9.4 Short-term accuracy and offset test. In the same container as the Working Standard Electrode, the product form electrode(s) that is to be tested shall be immersed in the 0.6 N NaCl solution (see 4.4.9.1). The product form electrode(s) shall be allowed to equilibrate in the test solution for a minimum of 24 hours. Using measurement equipment as specified in 4.4.9, the differential potential between the Working Standard Electrode and each product form electrode(s) shall be measured to obtain the accuracy value and for comparison to the requirements in 3.4.4. The differential potentials between each product form reference cell shall be measured to obtain the offset of all tested product form electrodes and for comparison to the requirements in 3.4.4.

4.4.10 Long-term tests (performance B only).

4.4.10.1 Long-term test preparation. Only product form performance B cells shall require long-term accuracy and offset, static drift, temperature linearity, and water resistivity linearity testing.

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4.4.10.1.1 Standard reference cells and tank. Standard reference cells as defined in 4.4.9.3 shall be used to test the performance B product form cells during long-term testing. The long-term testing Standard reference cell test setup and tank shall meet the following criteria:

a. The Standard reference cells shall have a lead wire connected that is soldered to the cell's wire lead and fully potted to prevent saltwater intrusion into the connection.

b. A Standard reference cell tank with a minimum volume of 1 gallon (4 liters) shall be provided to accommodate the Standard reference cells while conducting the product form testing.

c. The Standard reference cell tank shall consist of a separate tank from the performance B reference cell test tank and shall contain 0.6 N NaCl solution.

d. The resistivity of the solution in the Standard reference cell tank shall be 18±1 ohm-centimeters.

e. Resistivity shall be checked regularly during all product form long-term testing (minimum once weekly) and distilled water shall be added, as necessary, to maintain the required resistivity.

f. A minimum of 10 Standard cells shall be placed in the Standard reference cell tank for 96 hours at room temperature to equilibrate.

g. All air bubbles shall be removed from the reference cell surfaces after immersion.

h. A DC potential measurement shall be conducted at the end of the equilibration period to determine the potential difference between each pair of the 10 cells.

i. Three of the Standard cells measuring within ±0.5 mV DC of each other shall be identified to be used for further testing as the Select Standards.

j. One of the three Select Standard cells, called the Base Standard, shall be used to compare all product form cells during long-term testing.

k. The other Standard cells shall be maintained as spares in the event one of the Select Standards fails.

4.4.10.1.2 Base standard failure. If the Base Standard drifts more than ±1 mV DC from the initial differential reading between any of the other Select Standards, the Base Standard shall be replaced by one of the two Select Standards.

4.4.10.1.3 Long-term product form test tanks. There are two test tanks required for long-term testing. The test tanks and test medium shall meet the following criteria:

a. Artificial seawater in accordance with ASTM D1141 or natural seawater shall be used as the testing medium within each tank.

b. The test tank shall be filled with seawater (see 6.4.3) to a sufficient level to cover all of the product form reference electrodes to begin the conditioning period. An equivalent acclimation tank shall be filled with the same medium and shall be allowed to equilibrate for a minimum of 24 hours to ambient conditions.

c. The product form performance B reference cells shall be immersed in a test tank of sufficient size to contain a minimum volume of water of 0.65 cubic foot (18 liters) per reference cell to be tested.

d. The product form cells situated within the test tank shall have a minimum of 2 inches (5.08 centimeters) of water below the lowest reference cells and a minimum of 3 inches (7.62 centimeters) of water above the uppermost reference cells.

e. Sufficient space shall be provided to accommodate the minimum bend radius of the reference cell cable between the reference cell and the tank wall, with the element face in a vertical position.

f. A minimum of 8 inches (20.32 centimeters) between or in front of each reference cell shall be provided.

g. A transfer pump shall be used to transfer seawater to the test tank from the acclimation or mixing tank in less than 10 minutes.

h. The test tank and the acclimation tank seawater shall be changed out to a new electrolyte on a daily basis during both the conditioning period and test periods (excluding weekends).

i. The Standard reference cell tank and the test tank shall be electrochemically connected via a salt bridge containing 0.6 N NaCl solution (equivalent seawater conductivity).

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j. The temperature of the test tank shall be logged at the same interval as the test tank reference cell data (see 4.4.10.1.5). A temperature stainless steel thermistor shall be used with an operating temperature range of at least -139 through +158 °F (-95 through +70 °C) with a resolution of 0.02 °F at +77 °F (0.01 °C at +25 °C). The probe shall be immersed near the center of the tank both horizontally and vertically throughout the conditioning period and long-term testing.

4.4.10.1.4 Product form conditioning period. Product form cells shall be immersed in the test tank and the potential differentials recorded every hour for a 4-week conditioning period prior to the commencement of accuracy and offset (see 4.4.10.2), static drift (see 4.4.10.3), environmental temperature (see 4.4.10.5), and resistivity (see 4.4.10.6) testing.

4.4.10.1.5 Potential measurements. Initial potential differentials of the product form cells shall be recorded within 15 minutes of immersion at the start of the conditioning period. Differential potential measurements shall be measured and recorded between each product form reference electrode under test and the Base Standard electrode. These measurements shall be made automatically, with a data logger such as the Keithley 2700 Multimeter/Data Acquisition System, or equal, interfaced with a desktop computer through a General Purpose Interface Bus (GPIB).

4.4.10.1.6 Temperature corrections. Each differential potential shall be corrected for temperature fluctuation by normalizing the differential to 77 °F (25 °C) using the following equation:

$$E(25) = E(T) + 0.33 \times (25 - T)$$

Where:

$E(25)$ = Voltage potential differential at 25 °C in mV

$E(T)$ = Voltage potential differential at temperature T (as recorded) in mV

T = Temperature of water in Celsius at the time the potential differential was measured

For example, if the potential differential between a product form reference electrode and the Standard working electrode is 0.5 mV DC and the seawater temperature in the test tank is measured as 23 °C, then:

$$E(25) = 0.5 + 0.33 \times (25 - 23)$$
$$= 0.5 + 0.66 = 1.16\ mV\ DC$$

4.4.10.1.7 Offset (variance) recording values. The product form reference electrodes potential differentials between the elements and the Base Standard shall be documented at initial immersion, 24 hours, 2 weeks, and at the end of the 4-week conditioning period. In addition, the maximum offset (see 6.4.2) shall be calculated and recorded. For example, in table IV, at 24 hours, the difference between the highest element's potential differential (-2.37) and the lowest (-7.45) is 5.08 mV DC. At 4 weeks, this variance has been reduced to 1.66 mV DC.

TABLE IV. Example of potential differentials versus the base standard during conditioning.

Reference ID	Initial	24 hours	2 weeks	4 weeks
Element 1	-7.98	-7.45	0.63	1.06
Element 2	-2.36	-2.53	0.43	0.72
Element 3	-3.86	-3.53	-0.66	-0.60
Element 4	-1.60	-2.37	0.03	0.05
Element 5	-1.68	-2.56	-0.02	-0.16
Maximum offset	6.39	5.08	1.29	1.66

4.4.10.2 Accuracy and offset. At the end of the 4-week conditioning period, the differential potential between the product form cells and the Base Standard cell shall be measured and recorded to determine the accuracy values of the product form cells. At the same time, the differential potential between each pair of the product form cells shall be measured and recorded to determine the offset value of each product form cell pair.

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4.4.10.2.1 Accuracy and offset data analysis. The accuracy and offset values shall be compared to the requirements in 3.4.4. If the product form reference electrodes do not meet accuracy or offset criteria, the cells shall be rejected and no further long-term testing is required.

4.4.10.3 Static and environmental drift. Static drift and environmental effects (see 6.4.1) are shown on figures 2 and 3.

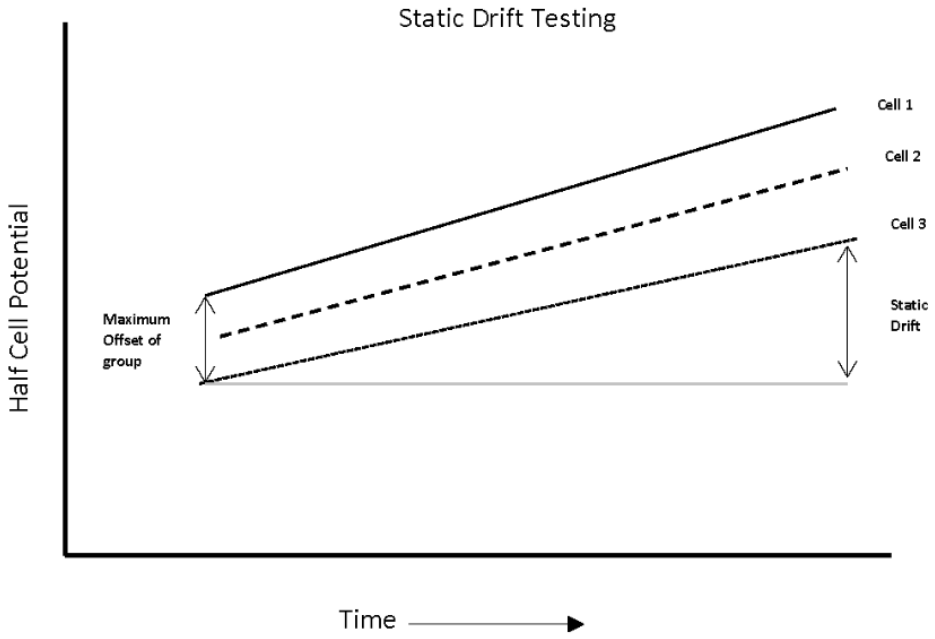


FIGURE 2. Static drift relevant to reference element testing.

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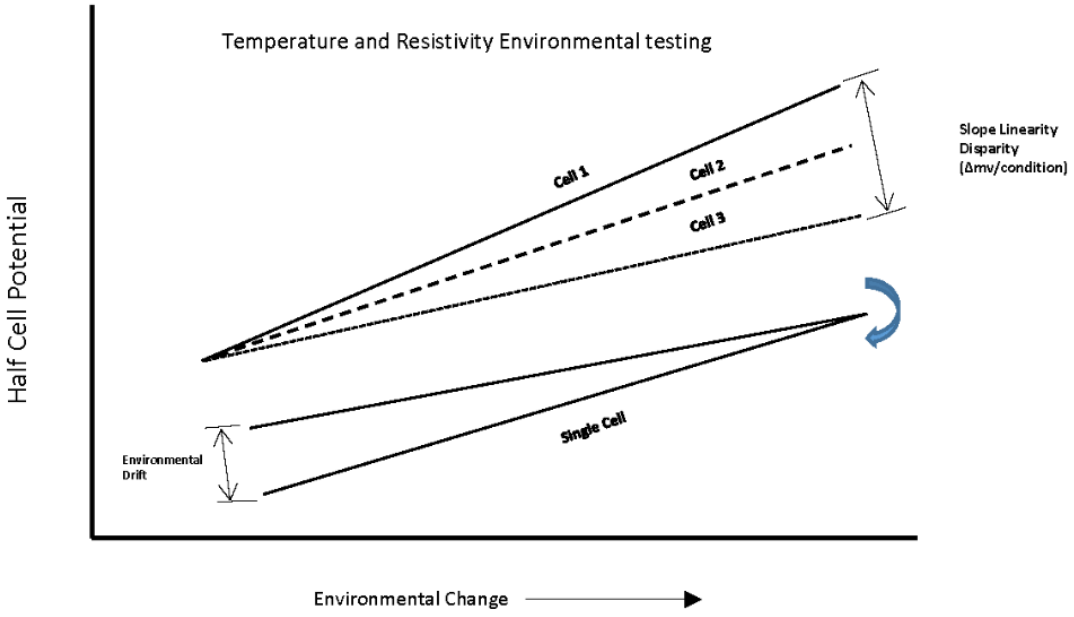


FIGURE 3. Environmental effects relevant to reference element testing.

4.4.10.4 Static drift test procedure. The 2-week period following the conditioning period is the static drift testing period. Data points shall be collected every hour for 2 weeks while changing the seawater daily as specified in 4.4.10.1.4 and applying the temperature correction as specified in 4.4.10.1.6. The data shall be plotted versus time as shown on figure 4.

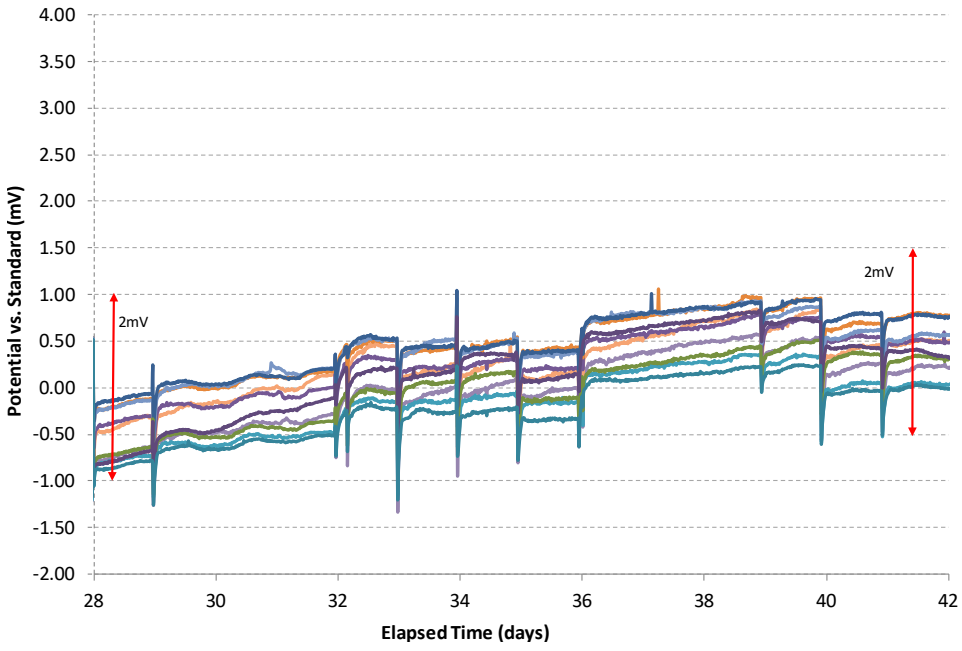


FIGURE 4. Plot of potential differentials normalized to 25 °C versus time.

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4.4.10.4.1 Static drift data analysis. The potential differentials at the start and end time of the 2-week static drift testing shall be recorded. The start and endpoint of the period shall be at approximately the same time of the day. Static drift shall be calculated by using the following equation:

$$StaticDrift_n = E_{final} - E_{initial}$$

The difference between the static drift for each pair of product form reference electrodes shall be calculated.

Example 1:

If the start value of the potential differential for reference cell 1 is -2.3 mV DC and the end value is -1.1 mV DC, the static drift is 1.2 mV DC. If the start value of the potential differential for reference cell 2 is -1.3 mV DC and the end value is 0.9 mV DC, the static drift is 2.2 mV DC. The difference in static drift is 1.0 mV DC and does not meet the criterion of less than or equal to 0.6 mV DC for passing, thus fails.

Example 2:

If the start value of the potential differential for reference cell 1 is -2.3 mV DC and the end value is -1.1 mV DC, the static drift is 1.2 mV DC. If the start value of the potential differential for reference cell 3 is -1.8 mV DC and the end value is -1.0 mV DC, the static drift is 0.8 mV DC. The difference in static drift is 0.4 mV DC and meets the criterion of less than or equal to 0.6 mV DC for passing.

4.4.10.5 Temperature linearity (environmental) testing. (see figure 3)

4.4.10.5.1 Temperature test setup. The test tank shall be filled with seawater. The seawater shall not be changed for the duration of temperature cycling. The temperature bath or heating/cooling element shall be set to a temperature that will increase the temperature in the test tank to a minimum of 90 °F (32.2 °C). The temperature shall not exceed 95 °F (35.0 °C). When the temperature has reached a minimum of 90 °F (32.2 °C), the temperature control element shall be reversed to begin cooling. The seawater temperature shall be cooled until it reaches a point below 40 °F (4.5 °C). When 40 °F (4.5 °C) has been reached, the temperature of the seawater in the test tank shall again be brought back up to 90 °F (32.2 °C). Following completion of a 90 to 40 °F (32.2 to 4.5 °C) and a 40 to 90 °F (4.5 to 32.2 °C) temperature cycle, the seawater shall be brought back to its initial temperature. The maximum temperature change in a 1-hour period shall not exceed 2 °F (1.1 °C).

4.4.10.5.2 Temperature linearity data analysis. The DC potential readings between the Standard Ag/AgCl cell and each of the product form cells shall be measured every 5 minutes. The potential readings shall be plotted versus temperature on a graph (see 3.4.6). A linear regression line shall be calculated from graphed data. The R-squared value (with a minimum of 20 data points) for a linear fit shall meet or exceed a value of 0.9800. While the slope of the line is dependent on the endpoints, the R-squared values will indicate any deviations from linearity between these two points (see figure 5).

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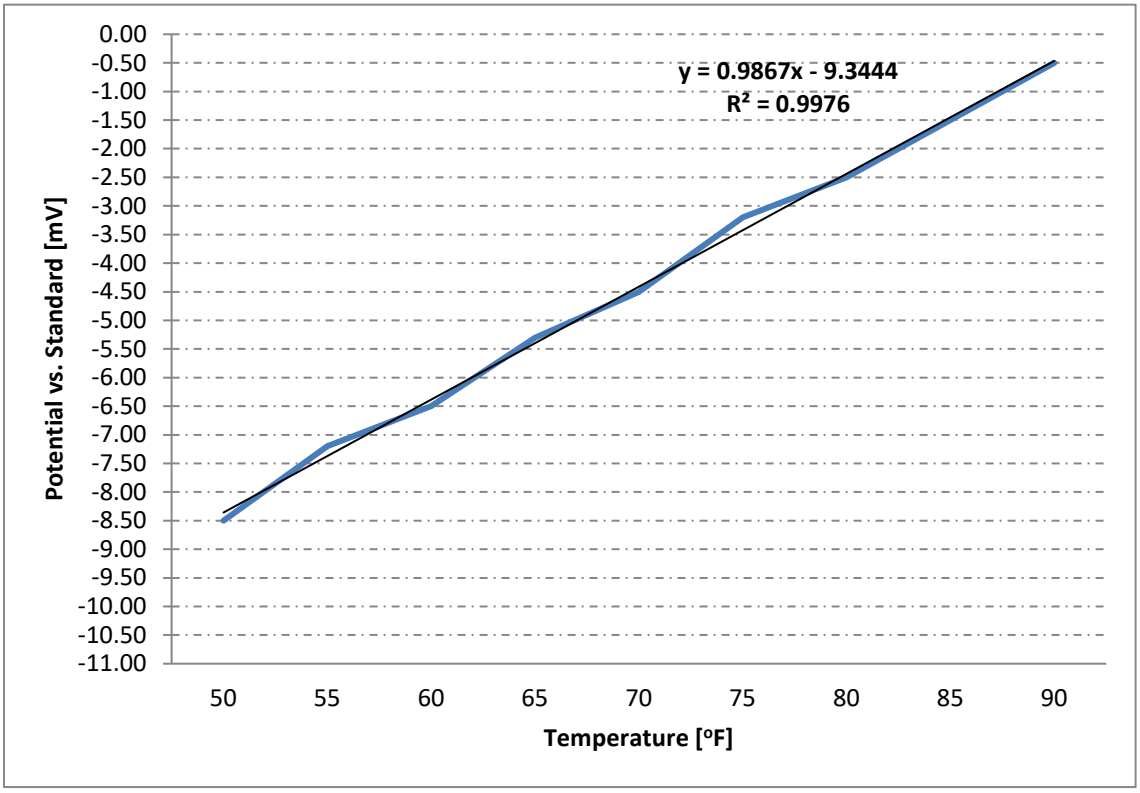


FIGURE 5. Acceptable deviation from linearity during temperature testing.

4.4.10.6 Water resistivity environmental linearity testing. (see figure 3)

4.4.10.6.1 Water resistivity test setup. The reference electrodes shall be cycled through seawater solutions of varying resistivities. Target resistivities are 18±1, 36±2, 72±5, 144±10, and 288±20 ohm-centimeters. Solutions of varying amounts of seawater and fresh water shall be mixed to achieve the targeted resistivities.

4.4.10.6.2 Water resistivity data analysis. The DC potential readings between the Standard Ag/AgCl cell and each of the product form cells shall be determined and plotted for each resistivity level. This value shall be measured at the same time index following the water change for each test solution resistivity. This time index shall be not less than 24 hours. Potential readings versus resistivity shall be graphed on a semi-log graph. A linear regression line shall be calculated from graphed data. The R-squared value shall meet the criteria specified in 3.4.7. Any deviation from linearity within the upper and lower limits of the test shall be immediate cause for rejection of the reference element.

4.4.11 Underwater replacement. The underwater replacement test requires one product reference cell assembly including the original element and one additional replacement element. The product assembly shall be placed with the original element in seawater for a period of 2 weeks. The reference cell potential versus the Standard shall be measured at a rate of every 30 minutes. After 2 weeks of exposure, the original reference element shall be replaced with a new element while underwater without removing the product assembly from the test tank. The potential shall be measured between the replacement element and the Standard for an additional 2 weeks. At the end of this measurement period, the potential difference measured shall be in accordance with 3.4.4 for performance A and B elements.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The reference cells are primarily intended for use in corrosion prevention in seawater of ship hulls, structures, sea chests, and similar marine structures. The reference cells supply a voltage signal to the automatic control ICCP system. Performance A and B reference cells procured to this specification are form, fit, and function replacements for prior revisions. The reference cells covered by this specification are military unique since they must be designed to meet the high level of shock and naval ship-specific pressure test requirements.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, grade, class, and performance level (see 1.2).

c. Requests to use sintering require NAVSEA approval (see 3.3.1.3).

d. Alternative surface area for a grade B element (see 3.3.1.4).

e. Ship-specific criteria for hydrostatic and hydrostatic pressure cycling (see 3.4.1 and 4.4.6.2).

f. When certification of chemical purity is to be provided (see 4.4.1).

g. When certification of thickness and longevity (life) calculations are to be provided in support of service life requirements (see 4.4.4).

h. When shock test procedure and test results are to be provided in accordance with MIL-DTL-901 (see 4.4.8).

i. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 23919 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Definitions.

6.4.1 Environmental effects. The change in reference cell potential (voltage) readings versus a standard as a function of environmental changes, such as temperature or seawater resistivity.

6.4.2 Maximum offset (variance). The difference between the highest and lowest potential (voltage) readings for each period.

6.4.3 Seawater. Artificial seawater in accordance with ASTM D1141 or natural seawater.

6.4.4 Static drift. The voltage differential between the initial and final potential values of a product form reference cell over a period of time in which there have been no environmental changes.

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6.5 Subject term (key word) listing.

Control

Corrosion

Electrode

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 2090-2019-002)

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